



Gennoor Tech
TRAIN. INNOVATE. BUILD.

5-DAY HANDS-ON BOOTCAMP · 40 HOURS

Production AI & MLOps

End-to-end ML lifecycle management for production systems

5 40 15+ 5+

DAYS HOURS HANDS-ON LABS PROJECTS

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Course Overview

This 5-day hands-on bootcamp covers everything from foundational concepts to advanced techniques, combining lectures, live demos, hands-on labs, and collaborative workshops for maximum learning impact.

Prerequisites: Python (intermediate), basic ML knowledge, Git, command line.

Who Is This For?



DevOps Engineers



ML Engineers



Platform Teams



L&D Teams
Internal trainers

Learning Outcomes

- 1 **Design end-to-end ML** — pipelines from data ingestion to deployment
- 2 **Implement CI/CD for** — ML using GitHub Actions and Azure DevOps
- 3 **Deploy models at** — scale with Docker, Kubernetes, and cloud endpoints
- 4 **Monitor production models** — for drift, degradation, and anomalies
- 5 **Version experiments (MLflow),** — datasets (DVC), and model registries

5-Day Curriculum

Interactive format with live demos, hands-on labs, and collaborative workshops. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

MLOps Foundations — Tracking & Versioning

HANDS-ON LABS + WORKSHOP

SESSION A

MLOps Foundations

- MLOps maturity levels (0→3) and organizational patterns
- ML lifecycle: data → train → validate → deploy → monitor → retrain
- MLflow: experiments, parameters, metrics, artifacts, model registry

SESSION B

Tracking & Versioning

- DVC for dataset versioning
- Lab: Track experiment lifecycle with MLflow + DVC

02

Containerization & Model Serving

DEMOS + LABS + DISCUSSION

SESSION A

Containerization

- Docker: images, containers, Dockerfiles, multi-stage builds
- Docker Compose for multi-service ML architectures
- FastAPI for inference endpoints, batch vs real-time patterns

SESSION B

Model Serving

- Model serialization: pickle, joblib, ONNX, TorchScript
- Lab: Production inference API with FastAPI + Docker

03

CI/CD for Machine Learning

LECTURE + HANDS-ON LABS

SESSION A

CI/CD for Machine Learning

- GitHub Actions: workflows, triggers, artifacts for ML
- Automated testing: unit tests, data validation, model quality gates
- Azure DevOps Pipelines for ML workloads

SESSION B

Hands-On & Practice

- Multi-environment deployment: dev → staging → production
- Lab: GitHub Actions pipeline that trains, tests, registers, deploys

04

Cloud MLOps — Azure, AWS & GCP

HANDS-ON LABS + WORKSHOP

SESSION A

Cloud MLOps

- Azure ML: workspaces, compute, managed endpoints
- AWS SageMaker: training jobs, model registry, endpoints
- GCP Vertex AI: pipelines, model registry, prediction

SESSION B

Azure, AWS & GCP

- Open-source stack: MLflow + Airflow + Kubernetes
- Lab: Deploy same model to Azure ML and SageMaker

05

Monitoring, Governance & Capstone

DEMOS + LABS + DISCUSSION

SESSION A

Monitoring, Governance

- Model monitoring: accuracy drift, data drift, concept drift
- Evidently AI and Weights & Biases for observability
- Automated retraining triggers and feedback loops

SESSION B

Capstone

- Architecture review: scalability, cost, security, compliance
- Capstone: End-to-end MLOps pipeline

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools demonstrated and practiced hands-on during the bootcamp

- Amazon SageMaker
- Docker
- FastAPI
- Google Vertex AI
- Kubernetes
- MLFlow

What You Take Home

 **MLOps Pipeline Templates (GitHub Actions, Azure DevOps)**
Included with course completion

 **Docker Starter Kit (Dockerfiles, Compose configs)**
Included with course completion

 **Monitoring Dashboard (Evidently AI templates)**
Included with course completion

 **MLOps Maturity Assessment Framework**
Included with course completion

Certification of Completion

 **Gennoor Tech — Certificate of Completion**
Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "Production AI & MLOps". Digitally signed by the Founder & Director. Validates proficiency in all course topics and hands-on competency.

Recommended Next Certifications

AI-103
Azure AI Solutions

AZ-400
Azure DevOps Solutions

DP-100
Azure Data Scientist

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited
Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of **16+ active Microsoft certifications** experience. Delivered AI and technology programs for Microsoft, EY, IBM, Boeing, including AB-100 (Agentic AI), AI-103, AZ-400, Saudi Aramco, HDFC Bank, and 80+ organizations across 6+ countries including the AI-300, PL-400, and GitHub certifications. 10-day AI Leadership program for MCIT Saudi Arabia.
Builder of open-source MCP servers and Copilot Studio integrations. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options ■ **On-Site** — customized at your office | 🖥️ **Virtual Live** — interactive via Teams/Zoom | 🌐 **International** — GCC, Africa, APAC, UK



Ready to Start Your Production AI & MLOps Journey?

Contact us to discuss a customized program
for your organization or register for the next open batch.

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